

# The blood

A PINHEAD-SIZED DROP OF BLOOD teems with as many as 6 million cells per cubic millimeter. No one saw these cells until the arrival of the microscope. Before that time, blood was believed to be deeply mysterious, and it was often worshiped and regarded as sacred. It was known to be the very essence of life – as it ebbed away, the body died. However, little was known of its contents. Physicians of ancient times tasted blood, found it sweet, and guessed it contained sugars. They saw it spurt bright red from cuts in arteries, and realized that it was under pressure. They watched it ooze dark reddish-blue from cuts in veins and wondered at the color change. Until the last century bloodletting was popular among doctors who were otherwise stuck for a treatment. It was not until about 1658 that Dutch researcher Jan Swammerdam gazed into his microscope and became the first person to identify and record individual blood cells. Thereafter, the microscope became a window into the blood's miniature world of red cells and white cells, platelets, and plasma, blood's liquid portion.



## BLOOD GROUPS

The reason why red cells may rupture or stick together in a transfusion was unraveled by Austrian scientist Karl Landsteiner (1868-1943) and his colleagues. In certain cases, chemicals in the plasma of one person's blood react with chemicals on the surfaces of the red cells (p. 13) in the other person's blood. The work led to the discovery of the ABO blood-group system in the early 1900s, then the M, N, and P systems in 1927. In 1940, the rhesus system was first detected in rhesus monkeys. Today, each person's blood is tested, and the transfused blood is selected and matched for maximum safety of the recipient. Landsteiner received the Nobel Prize for Physiology or Medicine in 1930.



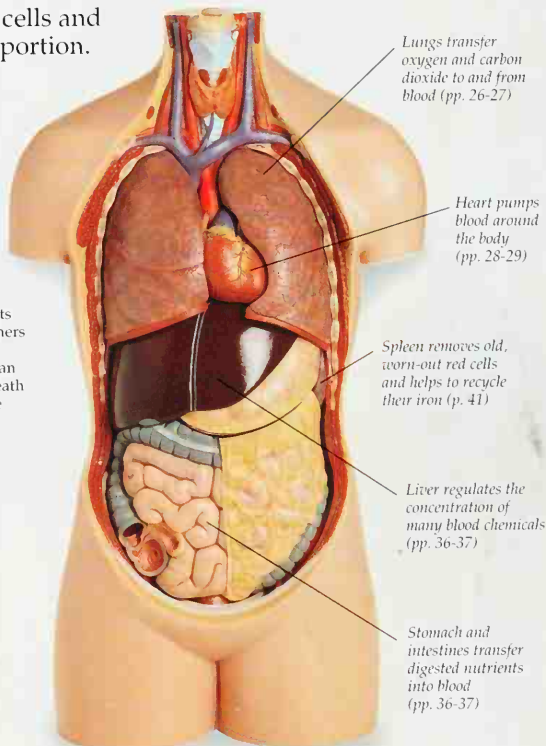
## BLOOD TRANSFUSIONS

Since antiquity, people have explored the notion that blood can be transfused (transferred) from one person to another – or even between animals and people, like this 17th-century transfusion from a dog. Some attempts were serious research, others were aimed at dramatic life saving. More often than not, the blood brought death instead of survival, as the red cells stuck together or burst open.



## FEEDING ON BLEEDING

Mosquitoes, fleas, and lice have been sucking blood for millions of years. It is a complete food for them, but blood turns up in superstition, legends, and the occult around the world. Many people have believed that by consuming blood, human or animal, qualities like strength, wisdom, and courage would be passed on. In Bram Stoker's enduring tale of *Dracula* (1897), the Count sucks the blood of young women.



## THE ROLES OF BLOOD

Blood is chiefly used for transport. It fetches and carries in endless journeys around the body. Exchanges between blood and tissues only take place in capillaries, whose walls are so thin that substances can pass through them. Blood spreads another body commodity: heat. It takes warmth from busy organs and muscles to cooler parts, maintaining an even body temperature, and warming the skin and extremities.